Digital Egypt Pioneers

DEPI – Round 3

Egypt's Energy Balance

Driving the Future

Energy Balance KPIs Dashboard (1971-2018)

CLS CAI3_DAT2_G4
Group 1

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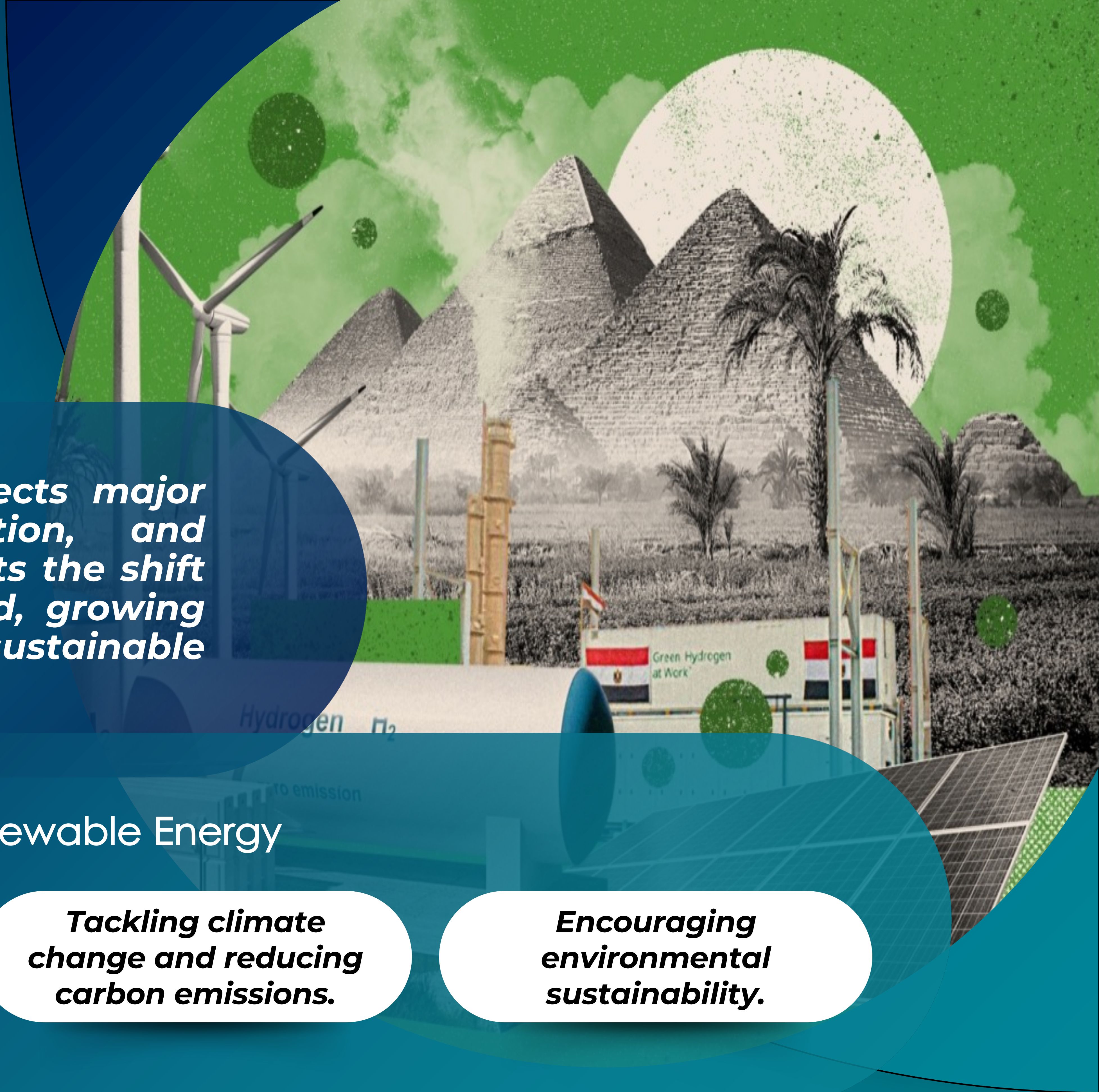
Project Management

Table of Content

Egypt Energy Balance (1971–2018)

Egypt's energy balance from 1971 to 2018 reflects major changes in energy production, consumption, and dependence on oil and gas. This period highlights the shift from domestic energy surplus to rising demand, growing imports, and the need for more efficient and sustainable energy management.



The Importance of Renewable Energy

- ***Reducing dependence on fossil fuels.***
- ***Tackling climate change and reducing carbon emissions.***
- ***Encouraging environmental sustainability.***

Egypt Vision 2030 for Sustainability

Egypt Vision 2030 places sustainability at the center of national development. This involves transitioning towards sustainability.

- It focuses on expanding clean energy, improving resource efficiency, and promoting responsible consumption to ensure long-term resilience.

- Across all sectors, including industry, agriculture, tourism, transportation, logistics, water management, and renewable energy.

2030
VISION OF EGYPT

Objective

This project aims to perform an in-depth data analysis of the Egyptian energy balance (1971-2018) by combining robust data analysis techniques with the sector-specific insights, providing a solid basis for evaluating the performance of the energy sector.

Scope

1. Assess Egypt status in relation to national and international commitments
2. Support the achievement of strategic and economic targets.
3. Ensure alignment with sustainability goals.
4. Contribute to strengthening energy security of Egypt

Scope

5 Encourage diversification of energy supply sources

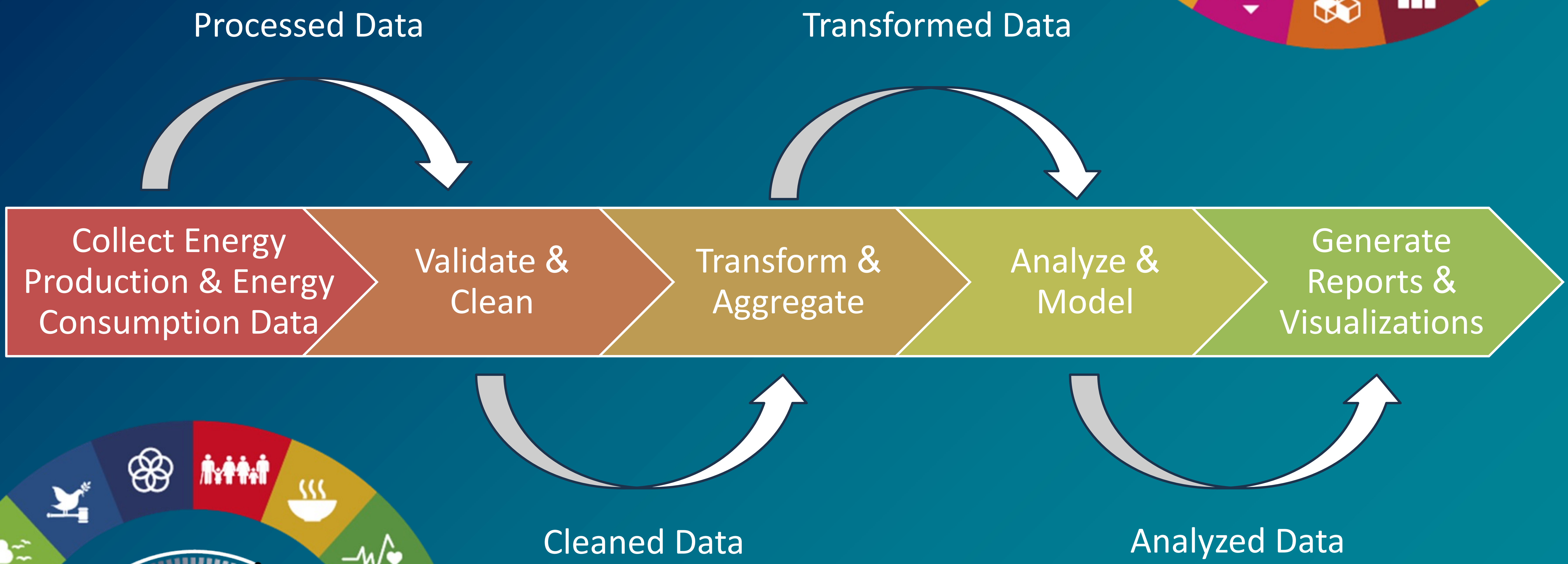
6 Promote energy efficiency and optimal resource utilization

7 Generate actionable knowledge to guide policy decisions, investment priorities, and long-term planning for a more sustainable and secure energy future

Project Timeline

TASKS	22 - 30 SEPTEMPER	1-10 OCTOBER	11-15 OCTOBER	16-20 OCTOBER	21-31 OCTOBER
TASK 1	Data Collection				
TASK 2		Data ETL			
TASK 3			Data relationship		
TASK 4				Dax measures	
TASK 5					Data Visualization

Data Flow Chart



Tools Used

MS Excel

Power Query

MS PowerBI

MS Powerpoint

Steps Applied on the Dataset

Data Collection

Extracted energy production & consumption data (1971–2018)

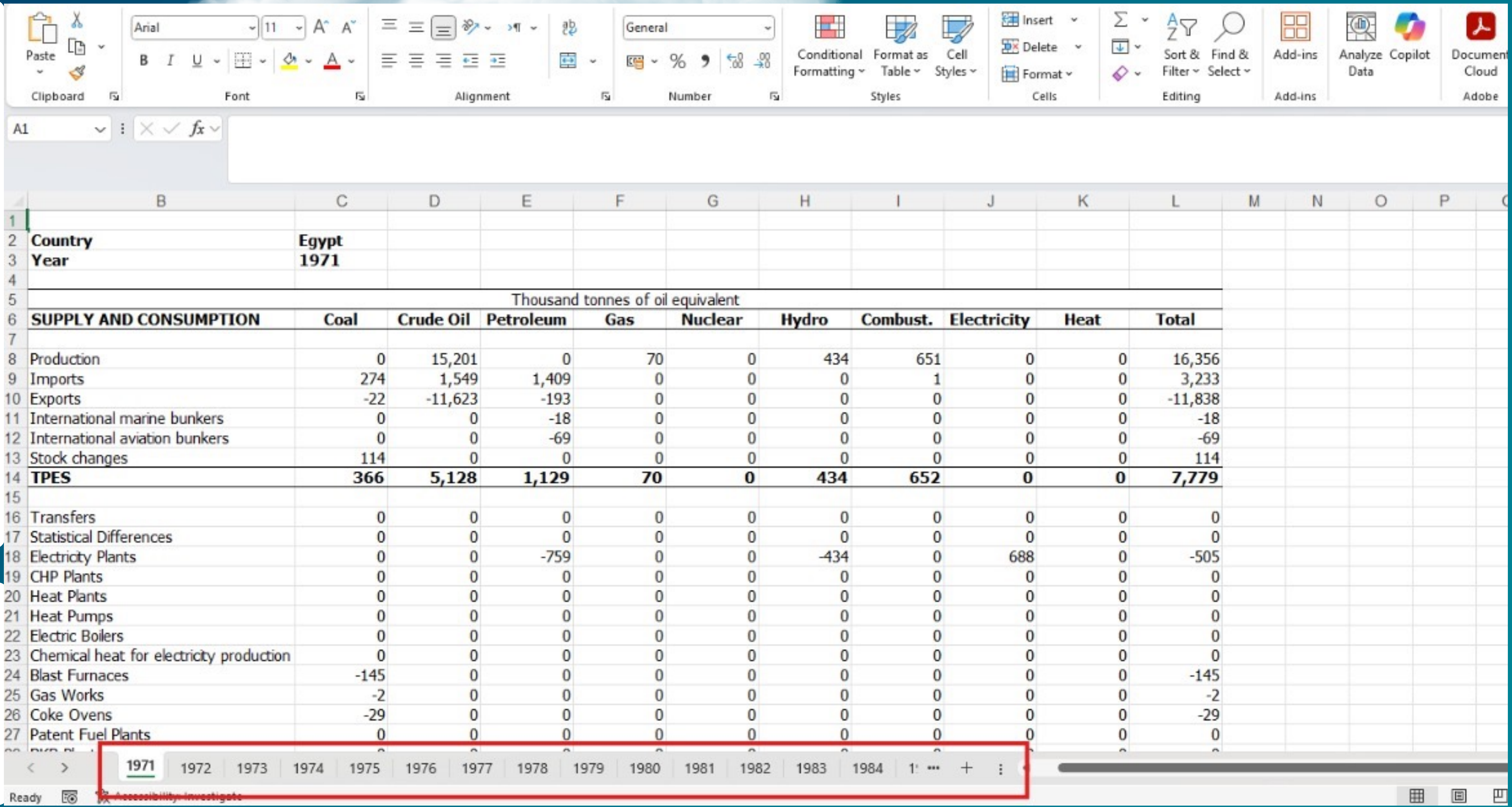
Sources: National energy databases (open source)

Data Collection

- ***Collect sheets from year 1971 to 2018 broken down for each year, organized into separate sheets***

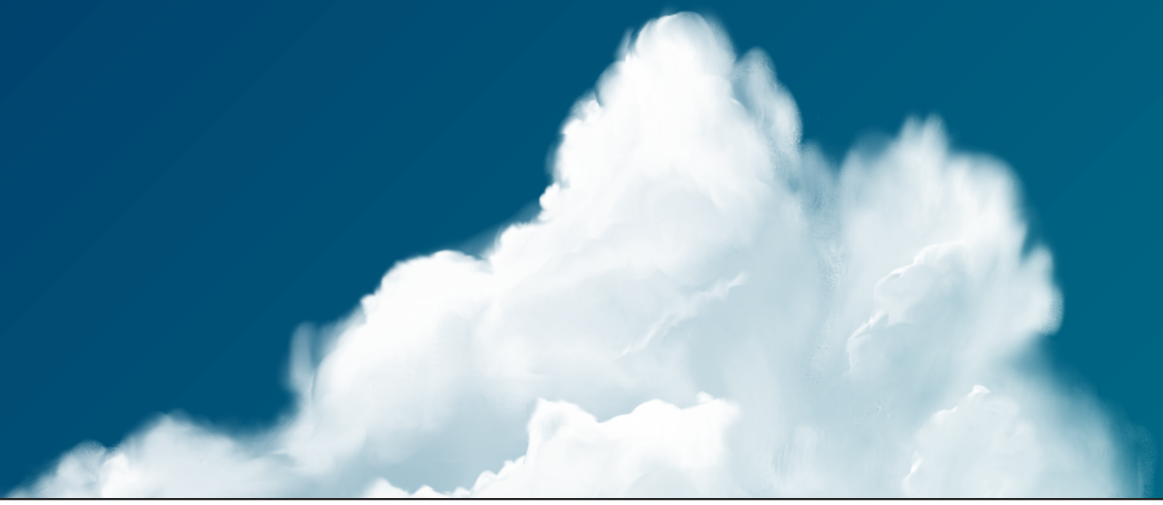
Program Used

Excel & Power Query



Country	**Egypt**									
Year	**1971**									
	Thousand tonnes of oil equivalent									
SUPPLY AND CONSUMPTION	**Coal**	**Crude Oil**	**Petroleum**	**Gas**	**Nuclear**	**Hydro**	**Combust.**	**Electricity**	**Heat**	**Total**
Production	0	15,201	0	70	0	434	651	0	0	16,356
Imports	274	1,549	1,409	0	0	0	1	0	0	3,233
Exports	-22	-11,623	-193	0	0	0	0	0	0	-11,838
International marine bunkers	0	0	-18	0	0	0	0	0	0	-18
International aviation bunkers	0	0	-69	0	0	0	0	0	0	-69
Stock changes	114	0	0	0	0	0	0	0	0	114
TPES	**366**	**5,128**	**1,129**	**70**	**0**	**434**	**652**	**0**	**0**	**7,779**
Transfers	0	0	0	0	0	0	0	0	0	0
Statistical Differences	0	0	0	0	0	0	0	0	0	0
Electricity Plants	0	0	-759	0	0	-434	0	688	0	-505
CHP Plants	0	0	0	0	0	0	0	0	0	0
Heat Plants	0	0	0	0	0	0	0	0	0	0
Heat Pumps	0	0	0	0	0	0	0	0	0	0
Electric Boilers	0	0	0	0	0	0	0	0	0	0
Chemical heat for electricity production	0	0	0	0	0	0	0	0	0	0
Blast Furnaces	-145	0	0	0	0	0	0	0	0	-145
Gas Works	-2	0	0	0	0	0	0	0	0	-2
Coke Ovens	-29	0	0	0	0	0	0	0	0	-29
Patent Fuel Plants	0	0	0	0	0	0	0	0	0	0

Data Collection



- Collect sheets of primary energy items, transformation, and consumption by sector, etc.

Program Used

Excel & Power Query

Year	SUPPLY AND CONSUMPTION	Coal	Crude Oil	Petroleum Products	Gas	Nuclear	Hydro Geotherm. Solar etc.	Combust. Renew. & Waste	Electricity
1971	Agriculture/Forestry	0	0	11	0	0	0	0	
1971	BKB Plants	0	0	0	0	0	0	0	
1971	Blast Furnaces	-145	0	0	0	0	0	0	
1971	Chemical and Petrochemical	0	0	0	0	0	0	0	
1971	Chemical heat for electricity producti	0	0	0	0	0	0	0	
1971	CHP Plants	0	0	0	0	0	0	0	
1971	Coke Ovens	-29	0	0	0	0	0	0	
1971	Commercial and Public Services	17	0	0	0	0	0	0	
1971	Construction	0	0	0	0	0	0	0	
1971	Distribution Losses	0	0	0	0	0	0	0	
1971	Domestic aviation	0	0	41	0	0	0	0	
1971	Domestic navigation	0	0	49	0	0	0	0	
1971	Electric Boilers	0	0	0	0	0	0	0	
1971	Exports	-22	-11623	-193	0	0	0	0	
1971	Fishing	0	0	0	0	0	0	0	
1971	Food and Tobacco	0	0	0	0	0	0	0	
1971	Gas Works	-2	0	0	0	0	0	0	
1971	Heat Plants	0	0	0	0	0	0	0	
1971	Heat Pumps	0	0	0	0	0	0	0	
1971	Imports	274	1549	1409	0	0	0	1	
1971	in Other Sectors	0	0	0	0	0	0	0	
1971	in Transport	0	0	0	0	0	0	0	
1971	INDUSTRY SECTOR	160	0	2591	0	0	0	331	
1971	International aviation bunkers	0	0	-69	0	0	0	0	
1971	International marine bunkers	0	0	-18	0	0	0	0	
1971	Iron and steel	158	0	0	0	0	0	0	
1971	Liquefaction Plants	0	0	0	0	0	0	0	
1971	Machinery	0	0	0	0	0	0	0	
1971	Mining and Quarrying	0	0	0	0	0	0	0	
1971	Non-ferrous metals	0	0	0	0	0	0	0	
1971	Non-metallic minerals	0	0	0	0	0	0	0	
1971	Non-specified (industry)	2	0	2186	0	0	0	331	
1971	Non-specified (other)	1	0	302	0	0	0	0	
1971	Non-specified (transport)	0	0	171	0	0	0	0	
1971	Oil Refineries	0	-5128	5091	0	0	0	0	
1971	Other Transformation	0	0	0	0	0	0	0	
1971	Own Use	0	0	-171	-70	0	0	0	
1971	Paper, Pulp and Printing	0	0	0	0	0	0	0	
1971	Patent Fuel Plants	0	0	0	0	0	0	0	
1971	Petrochemical Plants	0	0	0	0	0	0	0	
1971	Pipeline transport	0	0	0	0	0	0	0	
1971	Production	0	15201	0	70	0	434	651	
1971	Rail	0	0	0	0	0	0	0	
1971	Residential	12	0	1078	0	0	0	321	
1971	Road	0	0	1047	0	0	0	0	

Steps Applied on the Dataset

Data Collection

Extracted energy production & consumption data (1971–2018)

Sources: National energy databases (open source)

Data Cleaning

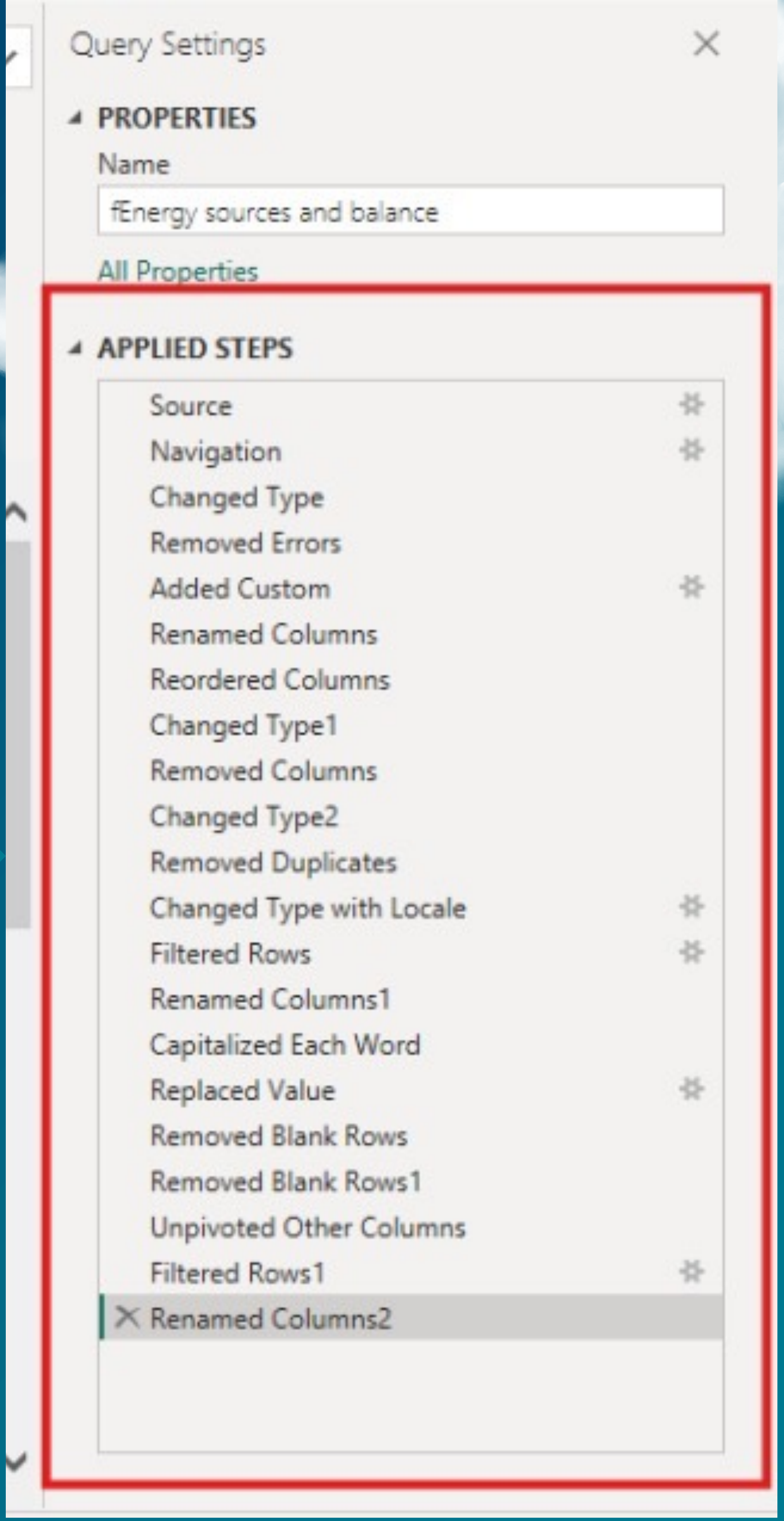
- *Handling missing or irrational values in specific years*
- *Standardizing product names & sector categories*
- *Correcting units and verifying yearly consistency*

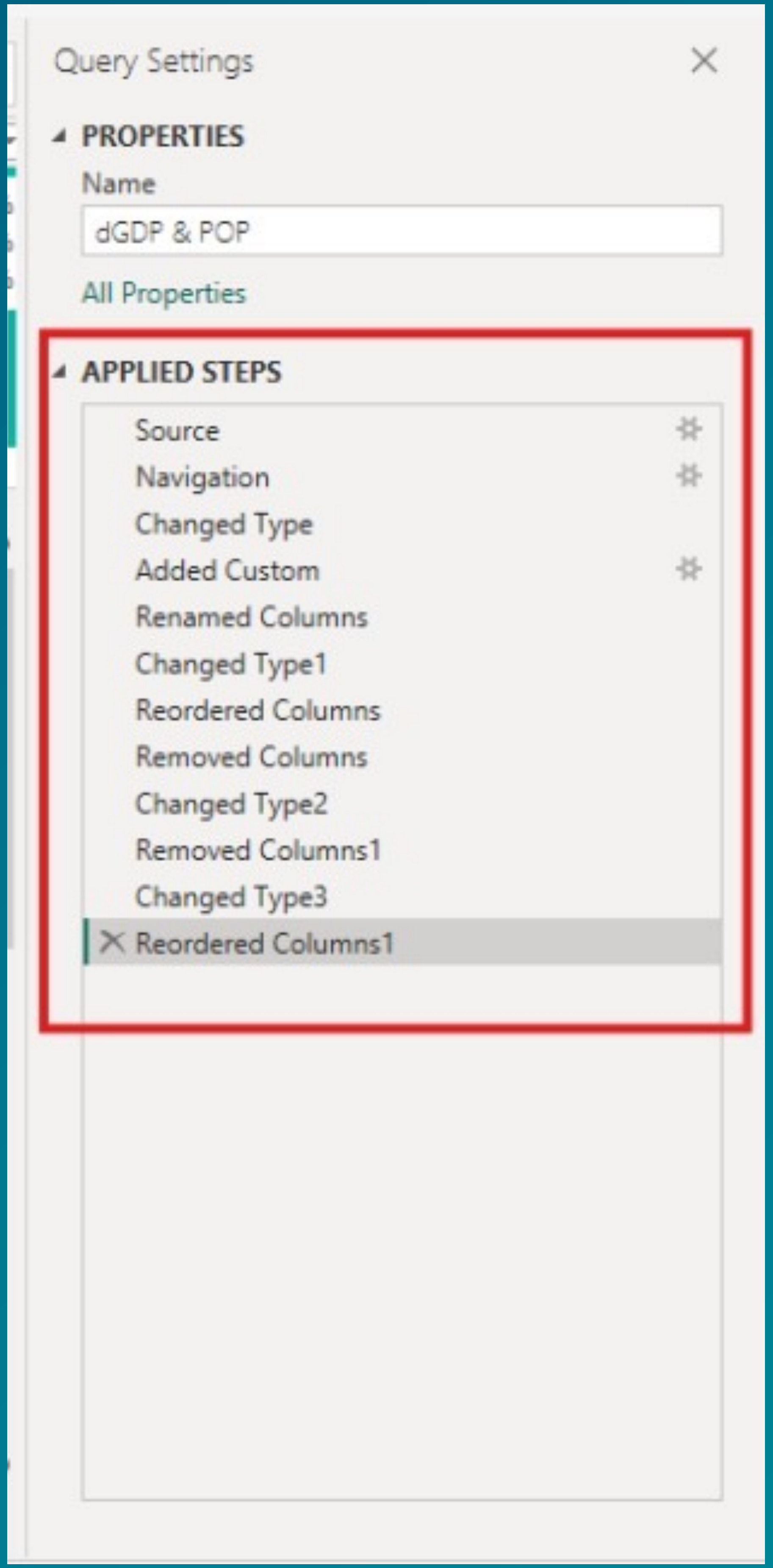
Data Cleaning

- *Normalize the fact table into dimension tables such as sectors and energy product types*
- *Clear empty cells*
- *Filled gaps using interpolation*
- *Removed invalid values*
- *Standardized units*

Program Used

Power Query





Steps Applied on the Dataset

Data Collection

Extracted energy production & consumption data (1971–2018)

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Data Cleaning

- Handling missing or irrational values in specific years
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- Correcting units and verifying yearly consistency

ETL & Data Modeling

- Transforming raw energy tables
- Merging data across energy sources and sectors
- Creating a star schema for clarity & performance

Data Transforming

- ***Normalization of data to reduce size and redundancy and enhance performance.***

Program Used

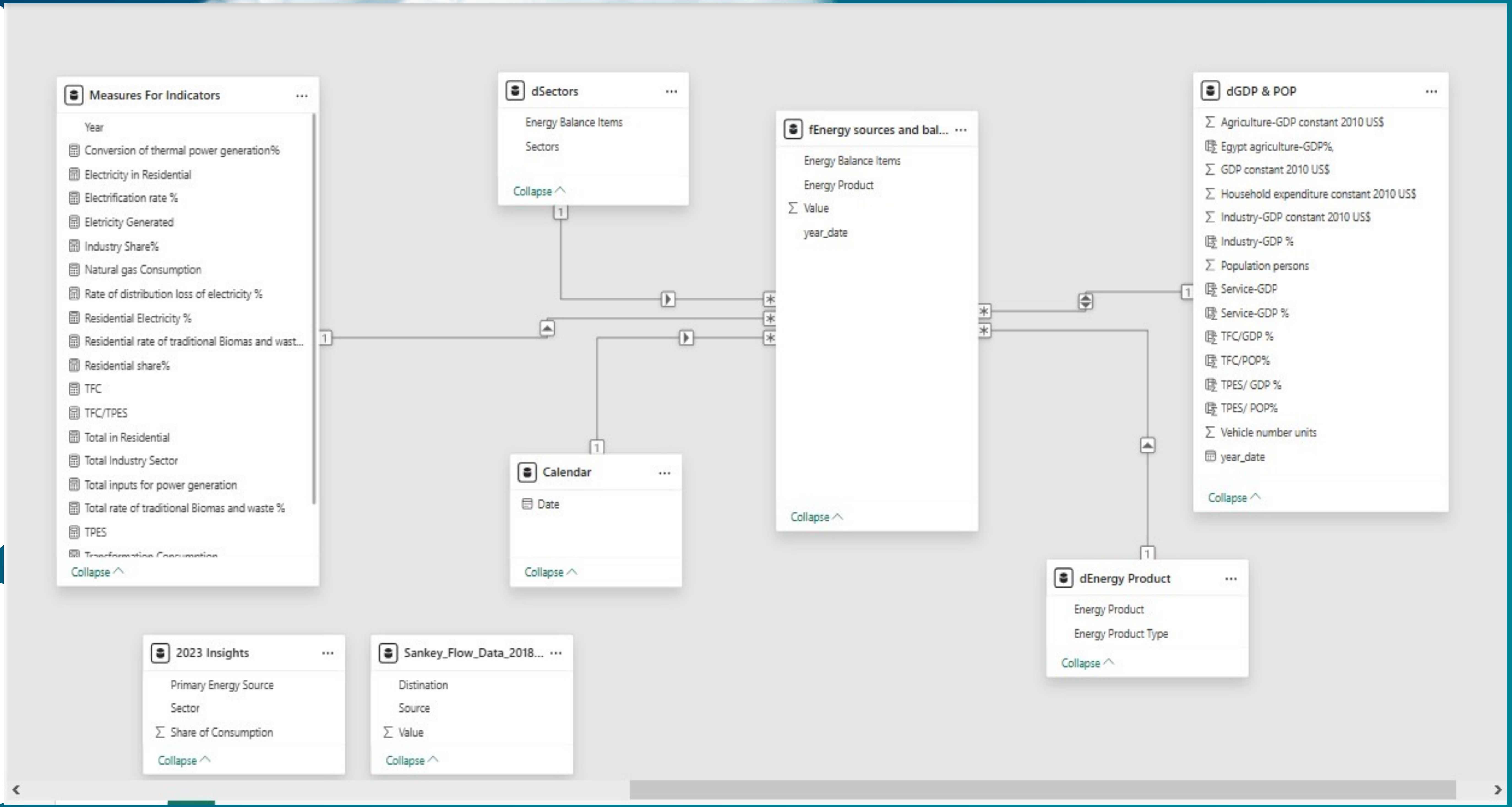
PowerBI

Energy Balance Items	Sectors
Iron and steel	Industry Sector
Chemical and Petrochemical	Industry Sector
Non-ferrous metals	Industry Sector
Non-metallic minerals	Industry Sector
Transport Equipment	Industry Sector
Machinery	Industry Sector
Mining and Quarrying	Industry Sector
Food and Tobacco	Industry Sector
Paper, Pulp and Printing	Industry Sector
Wood and Wood Products	Industry Sector
Construction	Industry Sector
Textile and Leather	Industry Sector
Non-specified (industry)	Industry Sector
World aviation bunkers	Transport Sector
Domestic aviation	Transport Sector
Road	Transport Sector
Rail	Transport Sector
Pipeline transport	Transport Sector
World marine bunkers	Transport Sector
Domestic navigation	Transport Sector
Non-specified (transport)	Transport Sector
Commercial and Public Services	Other Sectors
Fishing	Other Sectors
Non-specified (other)	Other Sectors
Residential	Residential
Agriculture/Forestry	Agriculture/Forestry

Energy Product	Energy Product Type
Coal	Non-Renwables
Crude Oil	Non-Renwables
Petroleum Products	Non-Renwables
Gas	Non-Renwables
Nuclear	Non-Renwables
Hydro Geotherm. Solar etc.	Renewables
Combust. Renew. & Waste	Renewables
Electricity	Non-Renwables
Heat	Non-Renwables

Data Modeling

- ***Tables are connected with primary key fields like year, energy product, and sector.***
- ***These relationships enable cross-filtering and dynamic visuals in Power BI.***



Program Used

PowerBI

Steps Applied on the Dataset

Measures Development

- KPI calculations (Energy Intensity, Renewable Share, TFC/GDP, Import Dependency...)
- Year-over-year comparisons
- Custom metrics

Measures Development

- ***Measures calculated with DAX for KPIs as (Electrification rate%, Electricity generated, TFC/TPES)***

Program Used

PowerBI

Measures For Indicators
- Conversion of thermal power generatio...
- Electricity in Residential
- Electrification rate %
- Eletricity Generated
- Industry Share%
- Natural gas Consumption
- Rate of distribution loss of electricity %
- Residential Electricity %
- Residential rate of traditional Biomas a...
- Residential share%
- TFC
- TFC/TPES
- Total in Residential
- Total Industry Sector
- Total inputs for power generation
- Total rate of traditional Biomas and was...
- TPES
- Transformation Consumption
- Transportaion share%
- Transportation Consumption(TOE)
- Year

Steps Applied on the Dataset

Measures Development

- KPI calculations (Energy Intensity, Renewable Share, TFC/GDP, Import Dependency...)
- Year-over-year comparisons
- Custom metrics

Data Analysis

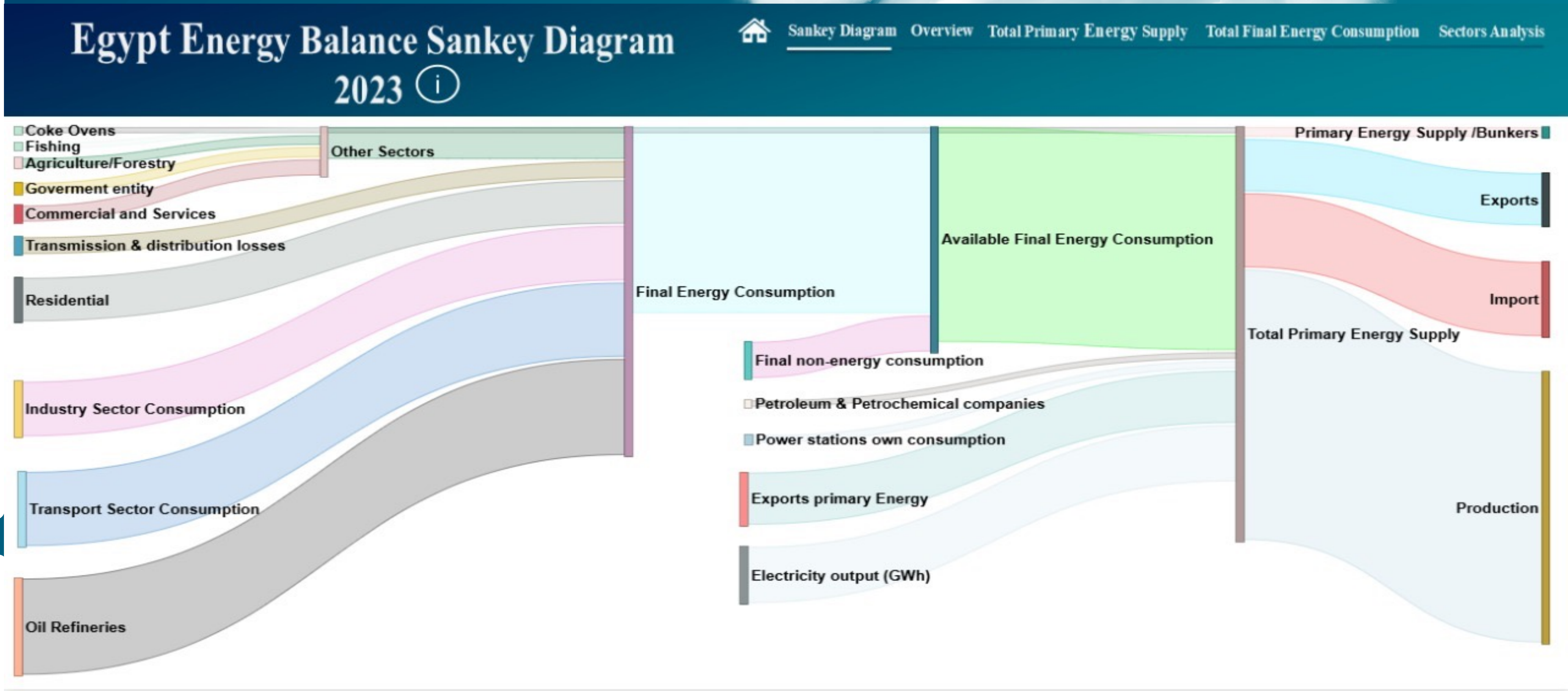
- Trend analysis of TPES, TFC, and electricity generation
- Sector consumption evaluation
- Efficiency and losses assessment
- Energy mix evolution analysis
- Renewable energy & sustainability indicators
- Identifying patterns, drivers, & key insights

Dashboard Visualization

- Multi-page Power BI report
- Unified color scheme + interactive navigation

Data Analysis & Visualization

- ***Build Supplementary page for clarification (Sankey Diagram)***

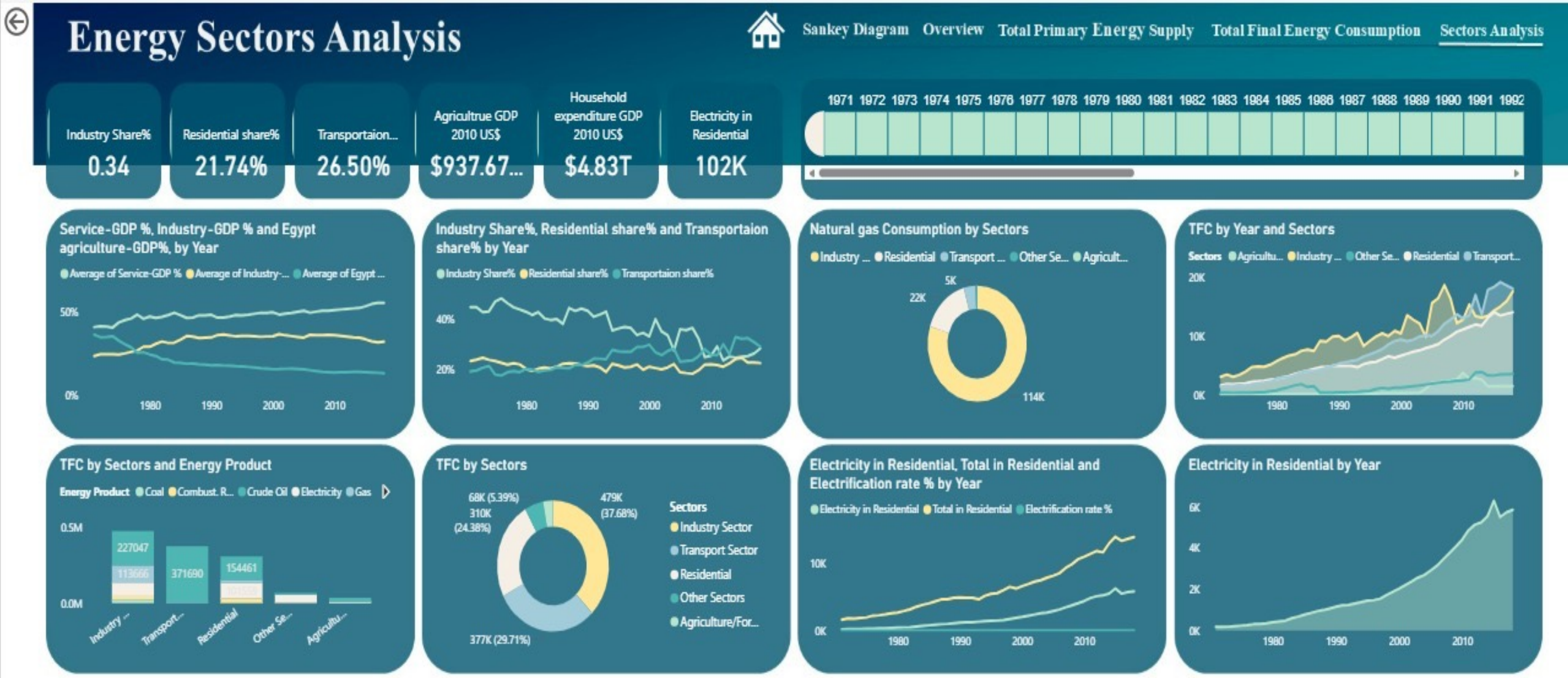


Program Used

PowerBI

Data Analysis & Visualization

- ***Build Overview with overall analysis (containing the most important features from all pages)***



Program Used

PowerBI

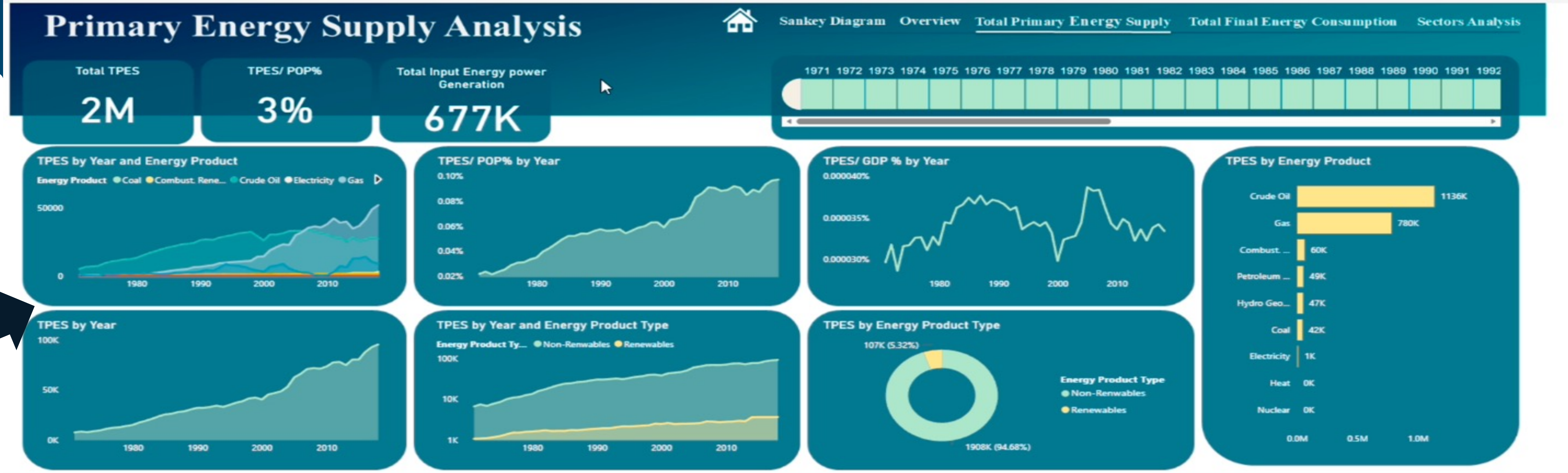
Data Analysis & Visualization

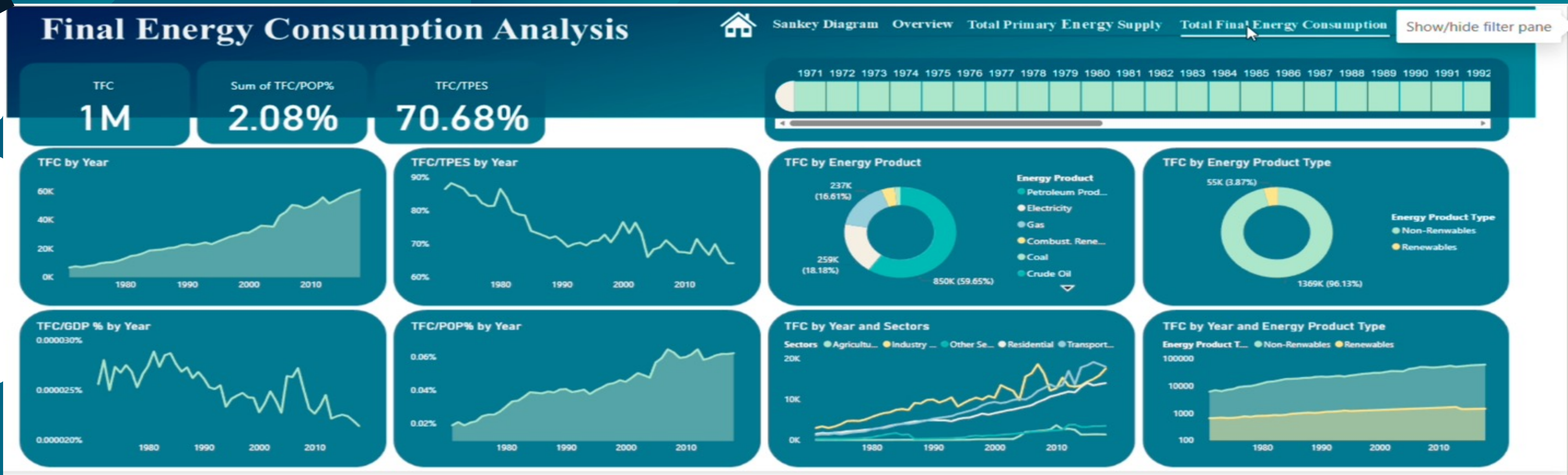
Build 3 main report pages:

- ***Total Primary Energy Supply***
- ***Total Final Energy Consumption***
- ***Sector Analysis***

Program Used

PowerBI





Data Analysis & Visualization

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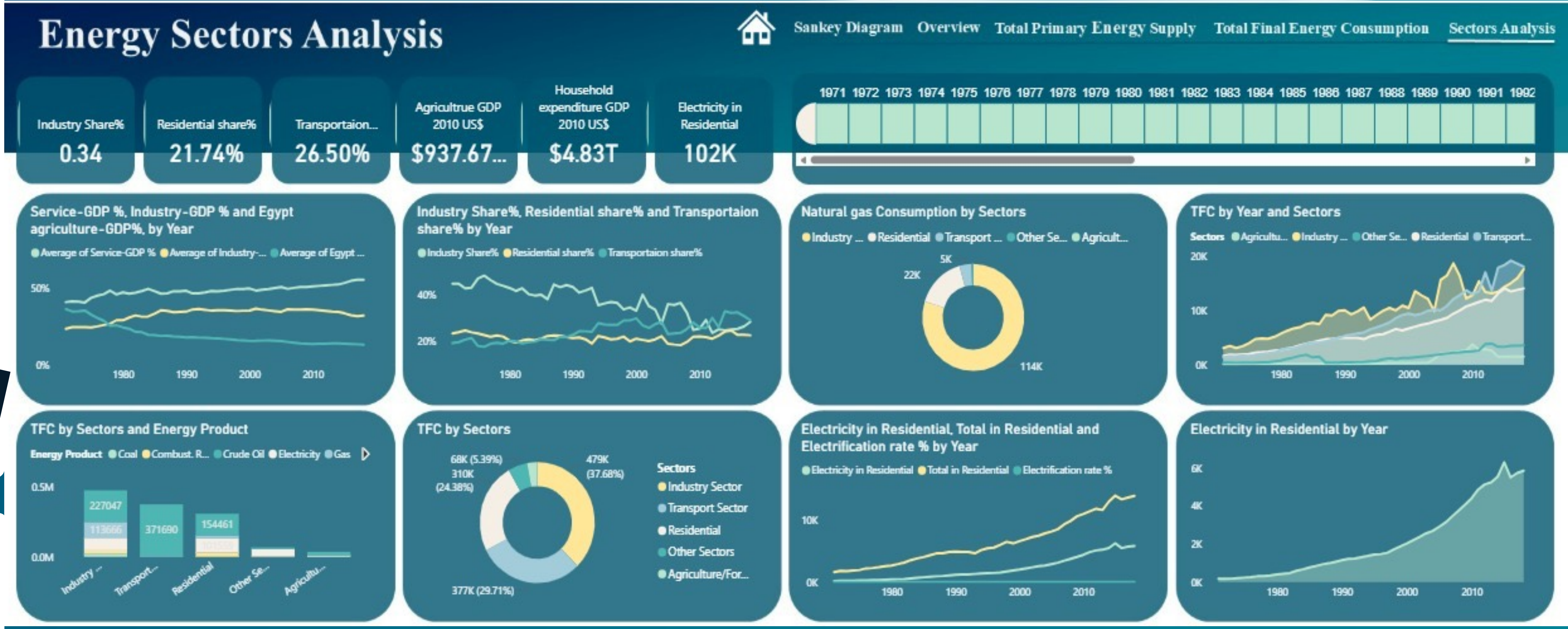
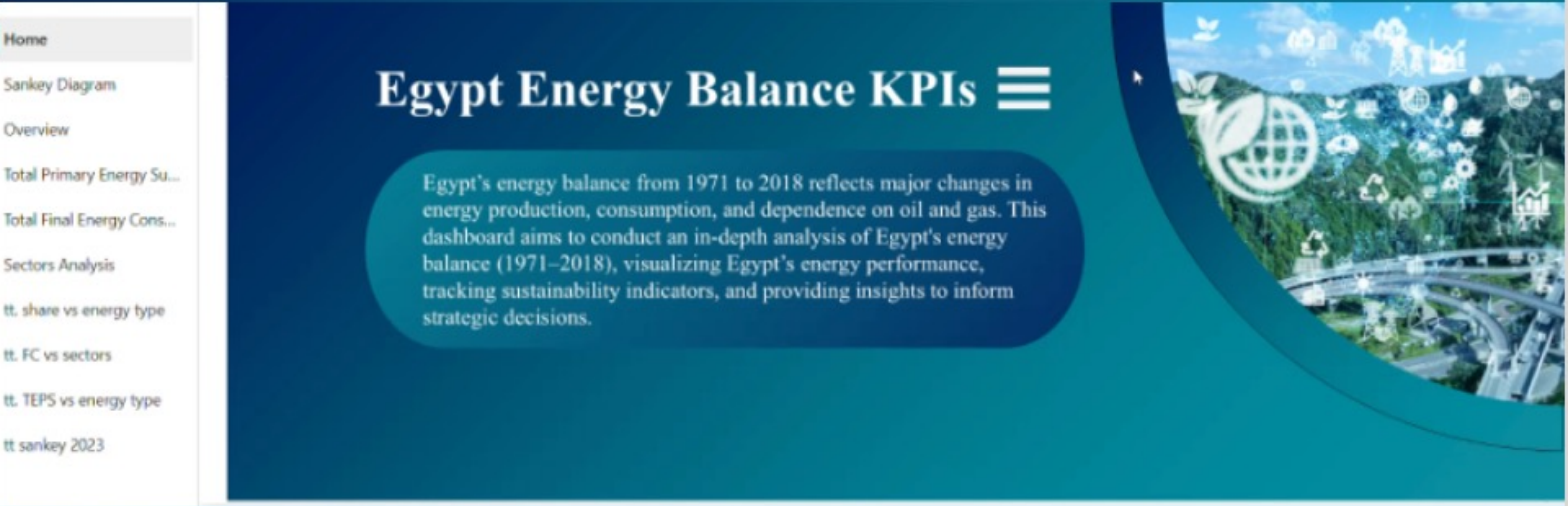


Program Used

PowerBI

Project Demonstration

Video Demonestration



Key Insights:

1 | Exponential Growth in Energy Demand

Observation

TPES, TFC, and electricity generation show continuous, steep growth over five decades.

Implication

- Power plants and grids face sustained pressure; massive investment is required just to keep up.

- Increasing risks to national energy security.

Recommendation

- Shift from *meeting demand* → *managing demand*.

- Implement strong energy efficiency programs, conservation policies & renewable projects across all sectors.

Key Insights:

2| Transport Becomes Largest Energy Consumer

Observation

Transport share rises steadily since the 1990s, surpassing industry as the top consuming sector.

Implication

- Transport becomes the largest sustainability and emissions challenge.
- Heavy dependence on petroleum fuels.

Recommendation

- Expand public transport (metro, electric trains).
- Promote electric vehicles (EVs).
- Introduce energy efficiency standards.

Key Insights:

3| Long-Term Decline in Energy System Efficiency

Observation

TFC/TPES ratio steadily declining → widening gap between supplied and consumed energy.

Implication

Large amounts of primary energy wasted → economic loss + higher environmental impact.

Recommendation

- Upgrade power generation technologies.

- Reduce transmission & distribution losses in the electricity grid.

Key Insights:

4| Economic Growth Has Not Decoupled from Energy Use

Observation

Energy intensity (TFC/GDP) does not show strong long-term decline.

Implication

- Future economic growth becomes increasingly costly and inefficient.
- Sustainability goals harder to achieve.

Recommendation

- Promote sustainability and knowledge-based sectors.
- Enforce strict energy efficiency codes for buildings and industries.

Key Insights:

5| Shift from Oil to Natural Gas Dominance

Observation

Natural gas becomes the primary energy source since early 2000s, replacing oil.

Implication

- Reduced reliance on imported oil.
- New dependency on a single fossil fuel (gas) → long-term supply and price risks.

Recommendation

- Transition from gas dependency to diversified renewable energy sources.
- Begin strategic shift before gas reserves decline.

Key Insights:

6| Renewable Energy Contribution Remains Very Low

Observation

Renewables consistently ~6% of TPES despite overall growth.

Implication

Missed opportunities for sustainability, economic savings, and green jobs.

Recommendation

- Set ambitious renewable energy targets.
- Provide investment incentives and streamline regulations.

Key Insights:

7| Heavy Reliance on Petroleum Products in Final Consumption

Observation

Petroleum fuels (diesel, gasoline) represent over 56% of TFC.

Implication

- High vulnerability to global oil price fluctuations.

- Major contributor to urban air pollution.

Recommendation

- Accelerate transport electrification.

- Expand electric public transit systems.

Key Insights:

8| Electrification & Gasification Improved Access but Hurt Efficiency

Observation

Sharp rise in final consumption of electricity and natural gas.

Implication

- Increased reliance on secondary energy → more conversion losses.
- Contributes to the long-term energy efficiency decline.

Recommendation

- Boost renewable electricity generation.
- Modernize gas and electricity infrastructure for higher energy efficiency.

Supervised by **Eng. Mahmoud Serag**

Team Members

Alaa Ragab Hani

Team Leader

- Presentation Design

- Documentation

Esraa Mohamed Hassan

- Data Cleaning & Engineer (ETL)

- Dashboard & Visualization Developer

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- Data Cleaning

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- DAX measures & calculation

- Dashboard & Visualization Developer

Nancy Gamal Eldin Azazy

Project Manager

- Data model Builder & Determining KPIs

- Data Cleaning

- Dashboard, Visualization & Sankey

THANKYOU!

Ready for Project Demonstration !